Artificial Intelligence in 3D Computer Graphics

Introduction

- Artificial Intelligence (AI) has become one of the most transformative technologies in modern computing.
- Computer graphics focuses on generating, manipulating, and rendering visual content.
- When combined, AI and computer graphics create faster workflows, higher-quality visuals, and new creative possibilities.

AI Matters in Graphics

- Analyze and generate visual data at high speed
- Automate complex or repetitive tasks
- Enhances realism and visual fidelity
- Expands what artists and designers can create

Evolution of AI in Graphics

- Early use: pattern recognition, image filtering, simple automation
- Deep learning revolution: neural networks capable of understanding and generating images
- Modern GPUs include dedicated AI hardware, enabling real-time intelligent graphics processing

Key Application Areas

- Image Generation

  AI models can create realistic images, textures, characters, and environments from scratch.

- Photo & Video Editing

  AI improves image quality, removes noise, enhances colors, and automates retouching.

- 3D Modeling & Animation

  AI assists with rigging, motion capture cleanup, and generating realistic animations.

- Rendering Acceleration

  AI reduces rendering time through denoising and super-resolution techniques.

- Virtual & Augmented Reality

  AI helps recognize objects, track movement, and optimize real-time rendering.

Image Generation

- AI models such as diffusion networks and GANs can produce photorealistic images
- Used in advertising, concept art, game design, and film
- Enables rapid prototyping and creative exploration

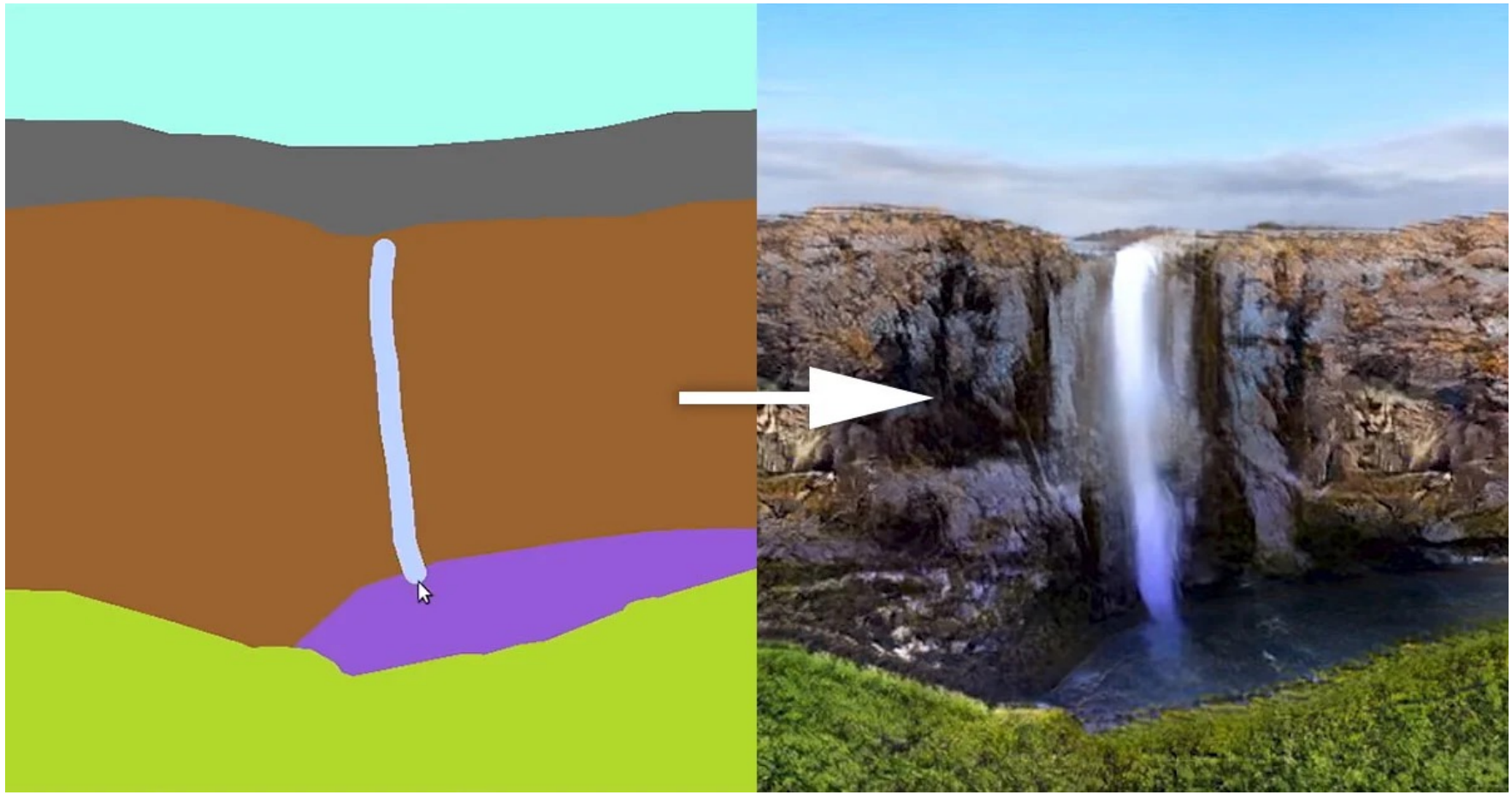
Examples - Image Generation

Midjourney / DALL·E / Stable Diffusion - Used to generate concept art, textures, characters, and environments from text prompts.



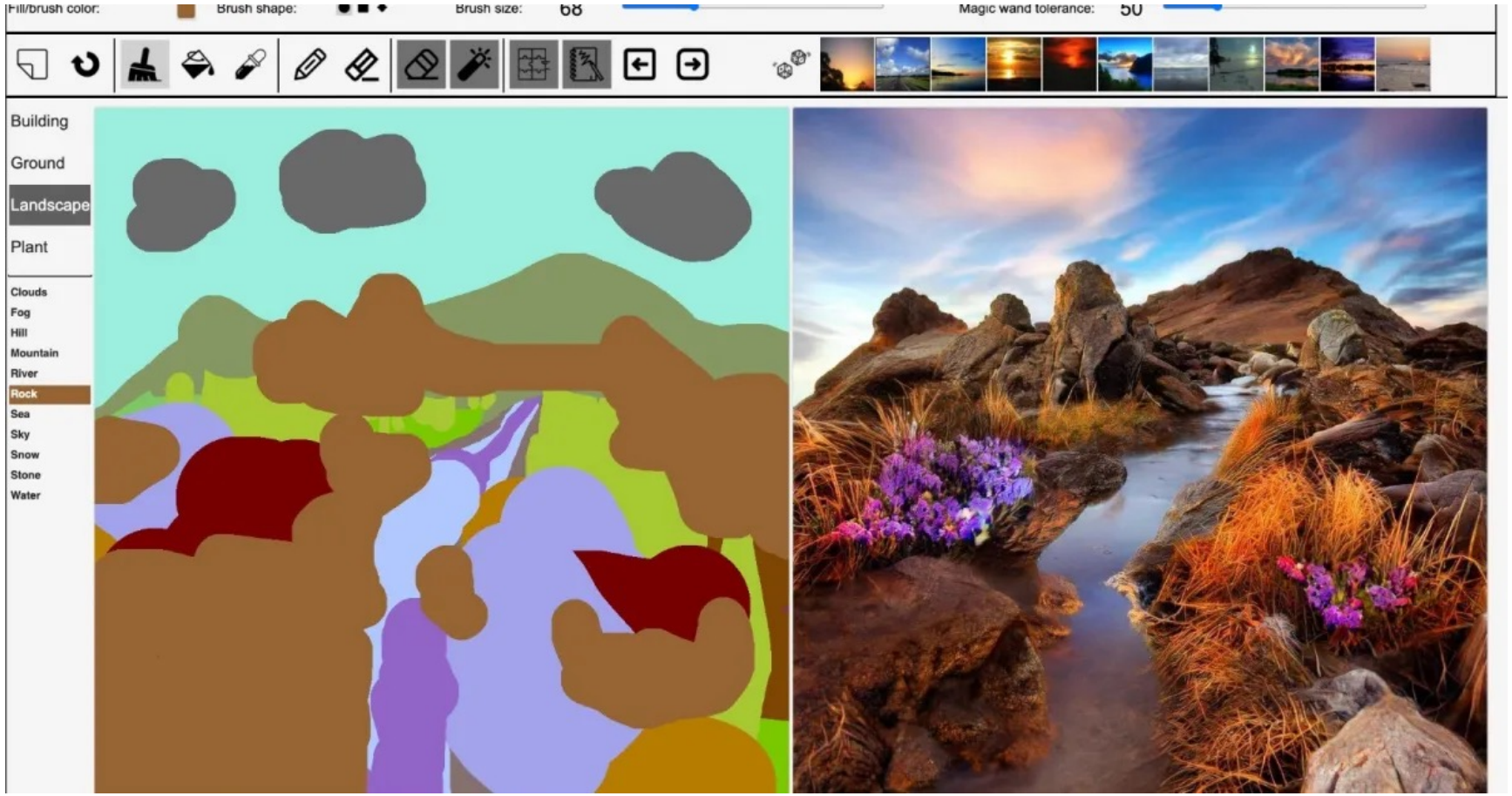
Examples - Image Generation

NVIDIA GauGAN - Turns simple sketches into photorealistic scenes using AI.



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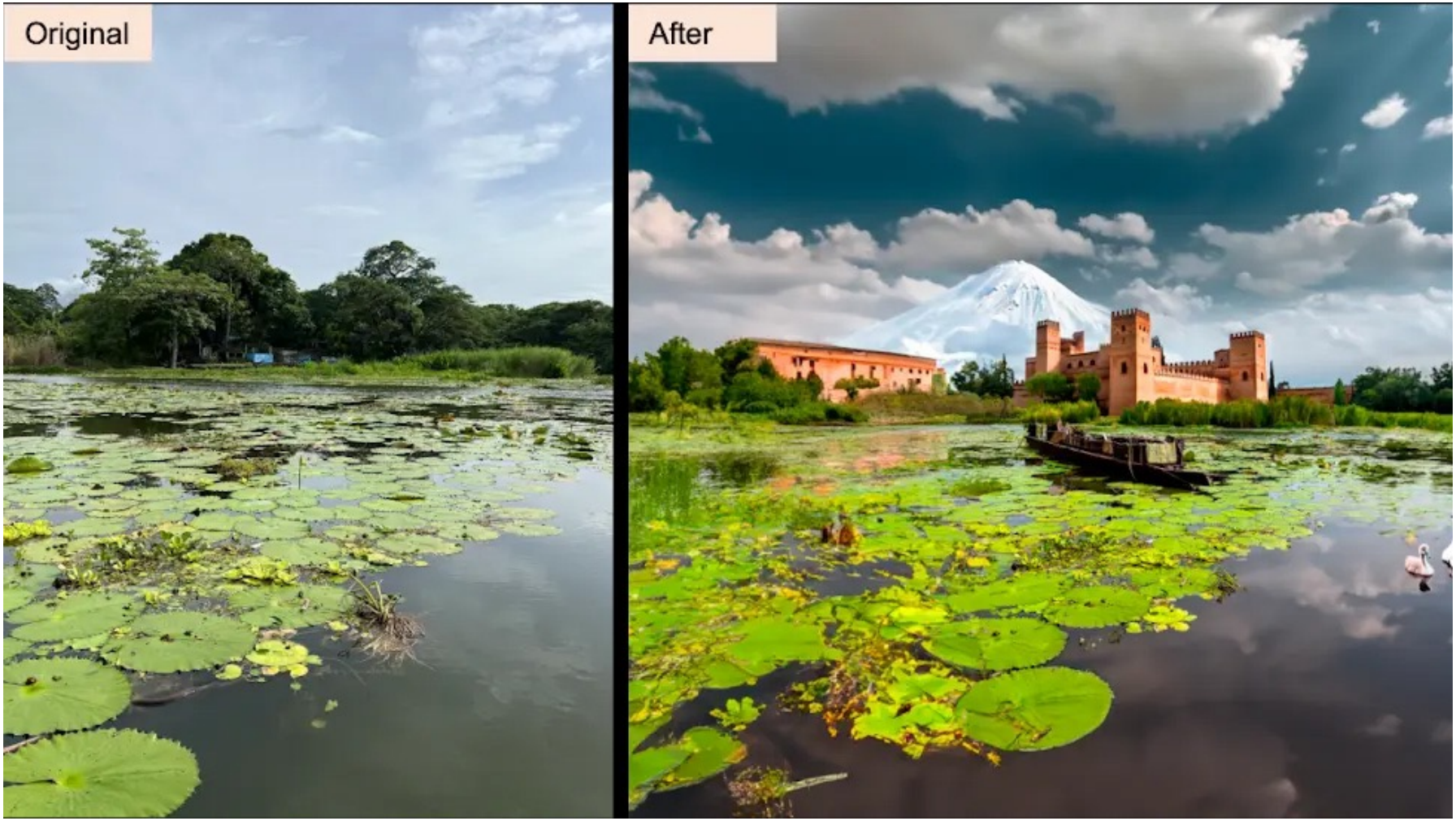


Intelligent Editing

- Automatic background removal
- Colorization of black-and-white images
- Noise reduction and sharpening
- Object detection for selective editing
- Smart video editing tools that identify scenes and suggest cuts

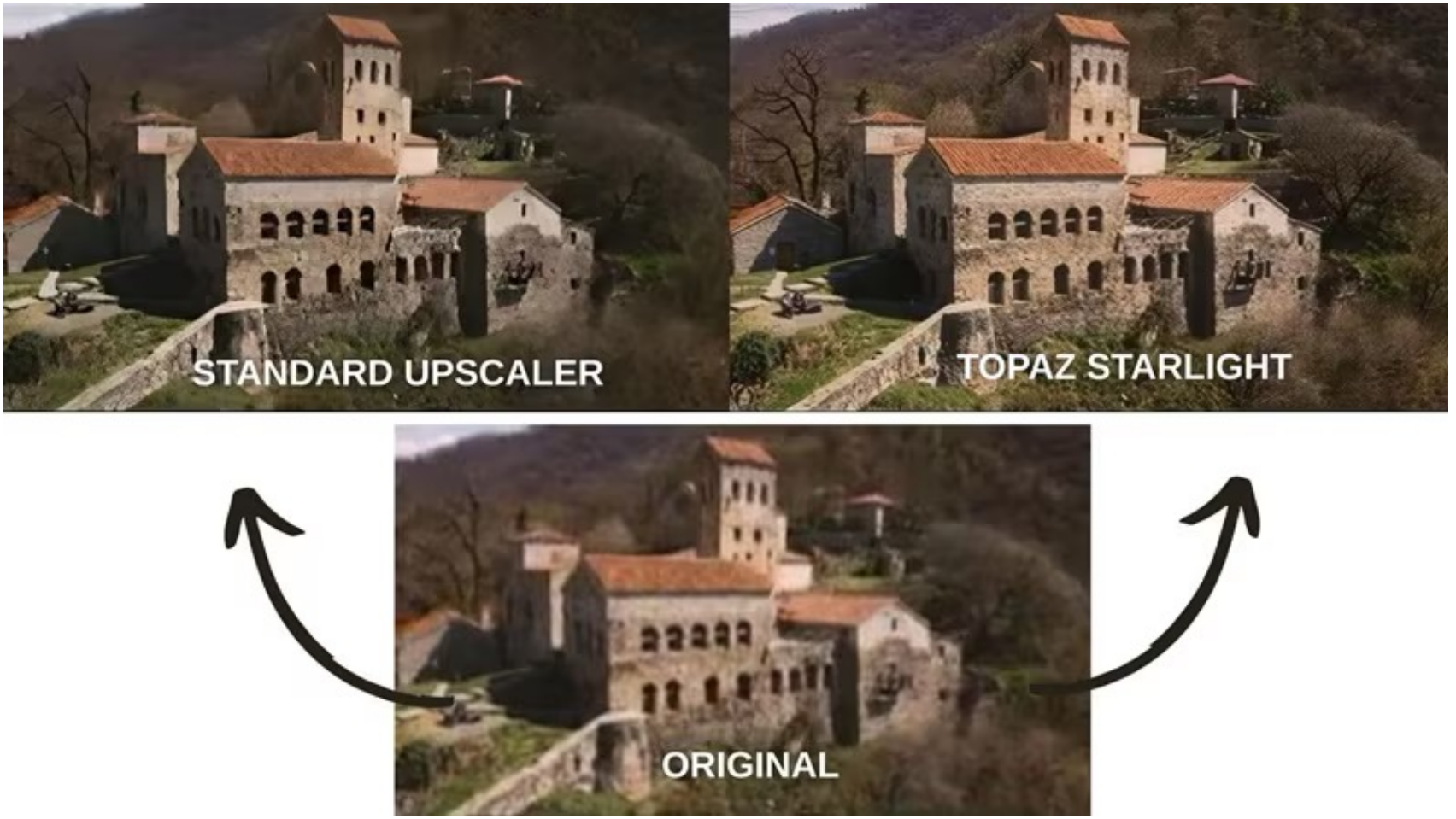
Examples - Intelligent Editing

Adobe Photoshop – Generative Fill - AI fills missing areas, removes objects, or extends images realistically.



Examples - Intelligent Editing

Topaz Video AI - Upscales low-resolution videos to 4K using AI super-resolution.



Examples - Intelligent Editing

Topaz Video AI - Restoring old footage for documentaries.



Examples - Intelligent Editing

DaVinci Resolve – Magic Mask - Automatically tracks and isolates people or objects in video.





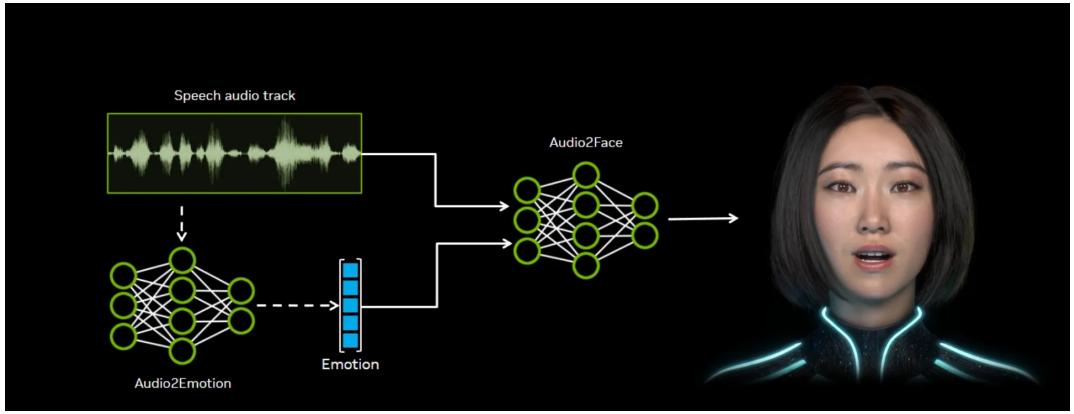


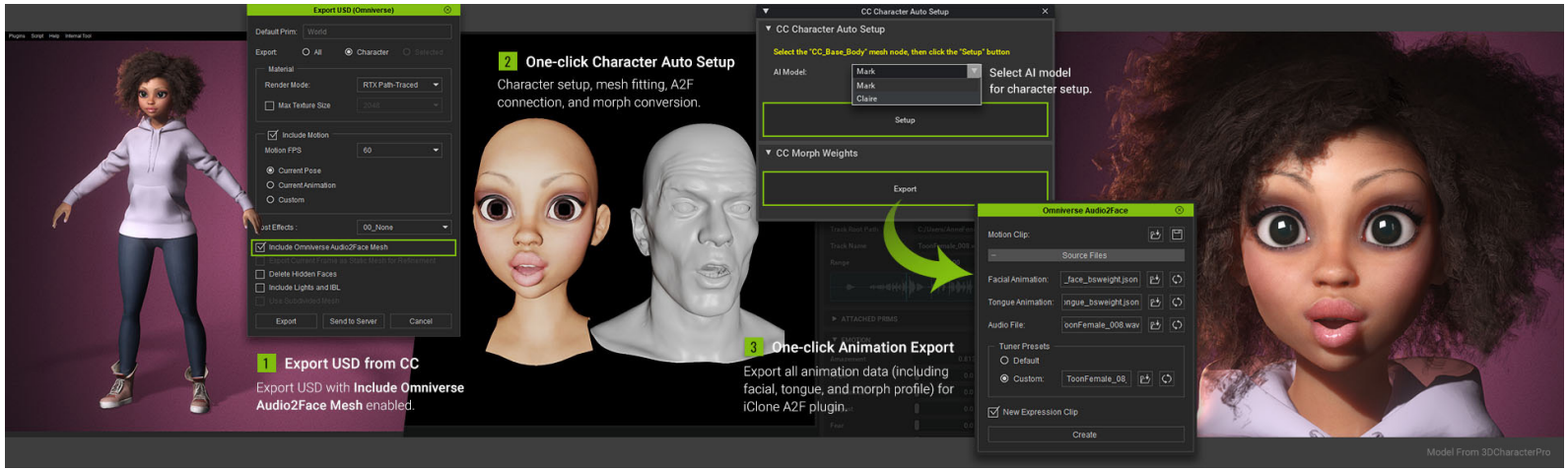
3D Modeling and Animation

- AI reconstructs 3D models from 2D images
- Automatic rigging for characters
- Motion prediction and smoothing for animation
- Simulation of physics-based effects (cloth, fluids, particles)

Examples - 3D Modeling and Animation

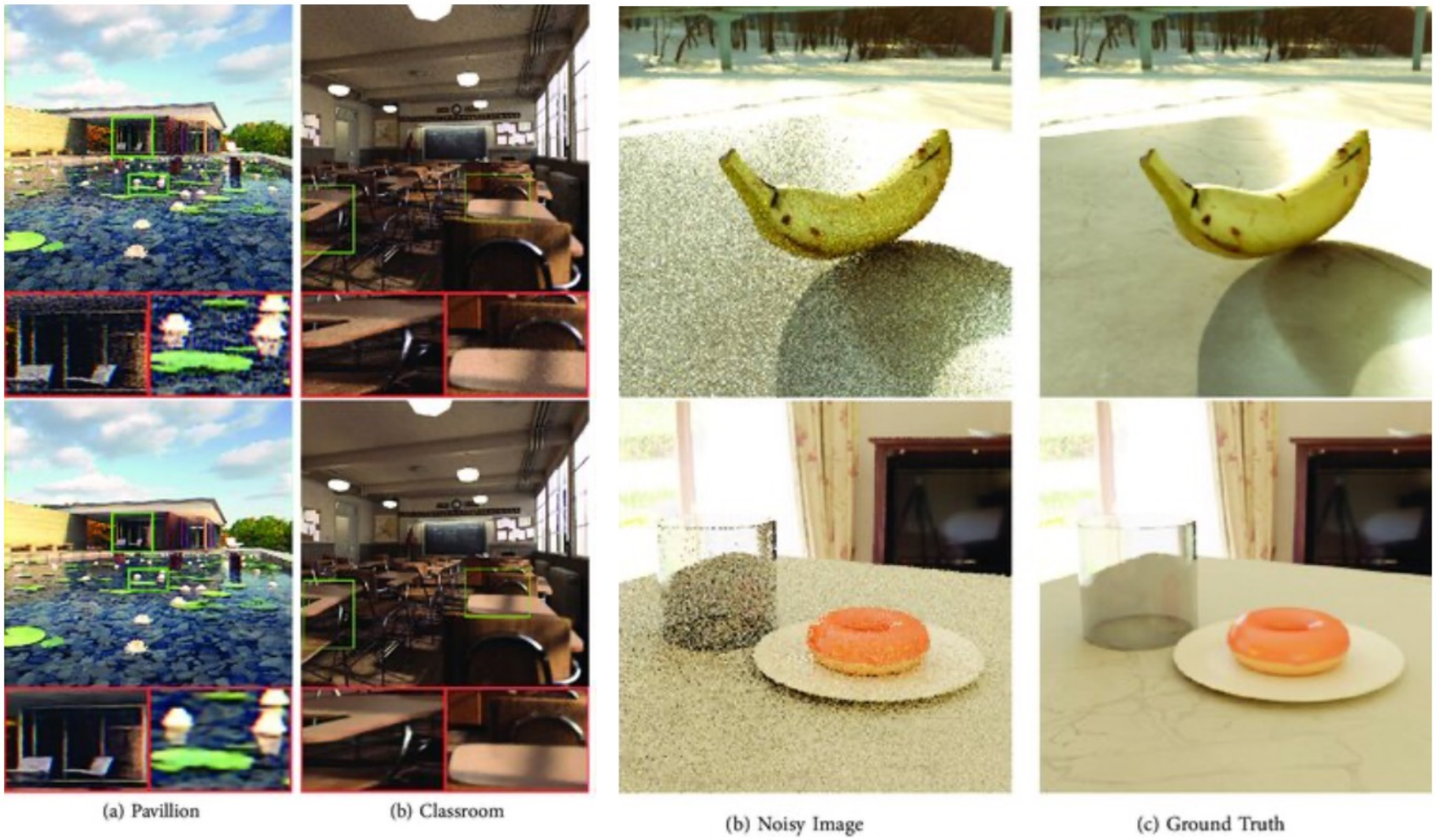
NVIDIA Omniverse Audio2Face - Generates facial animation from an audio file. A character's lips and expressions animate automatically based on dialogue.





Examples - 3D Modeling and Animation

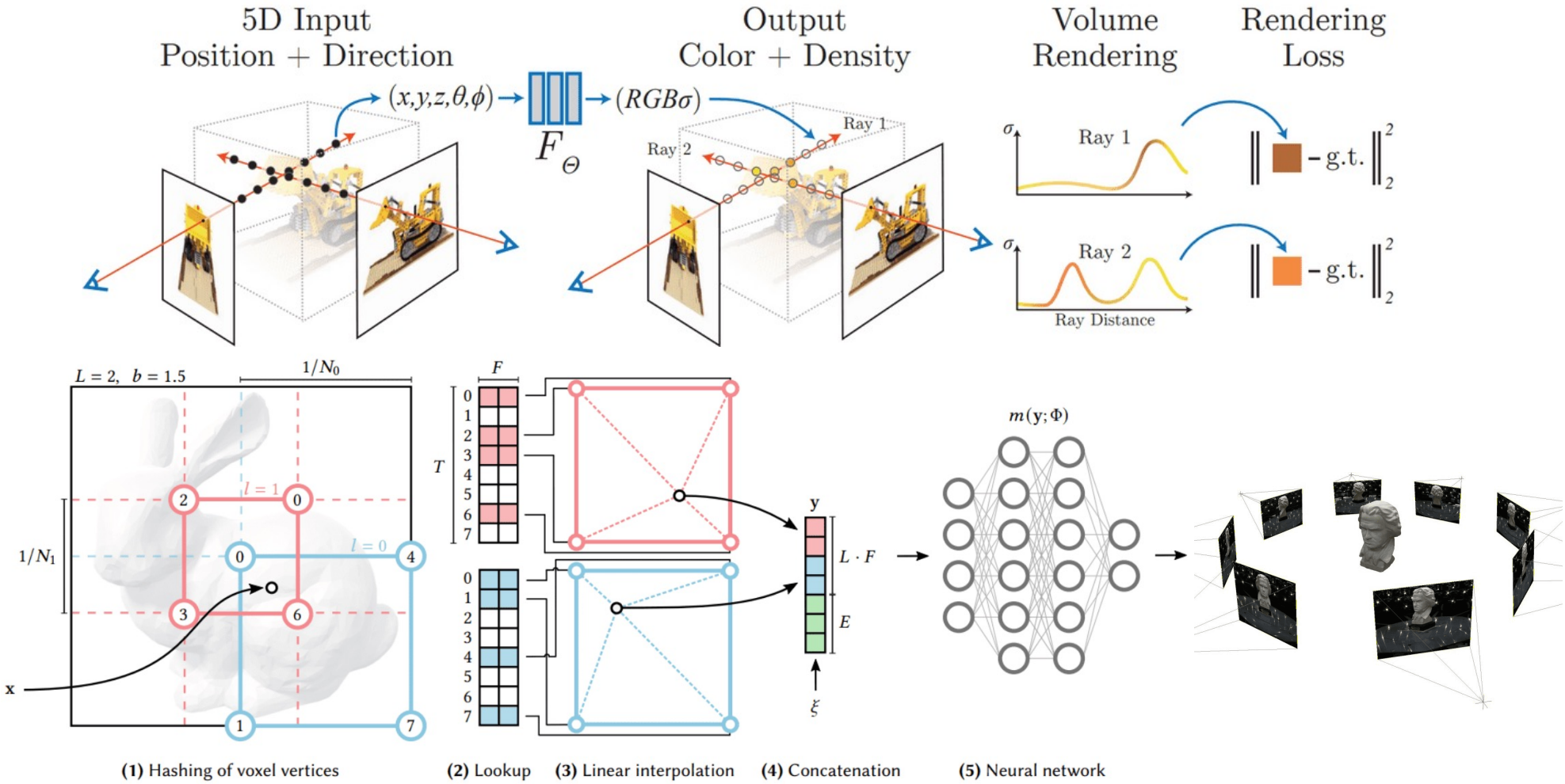
Blender – AI Denoising - Removes noise from 3D renders, reducing render time dramatically.



(a) Pavillion (b) Classroom (b) Noisy Image (c) Ground Truth

Examples - 3D Modeling and Animation

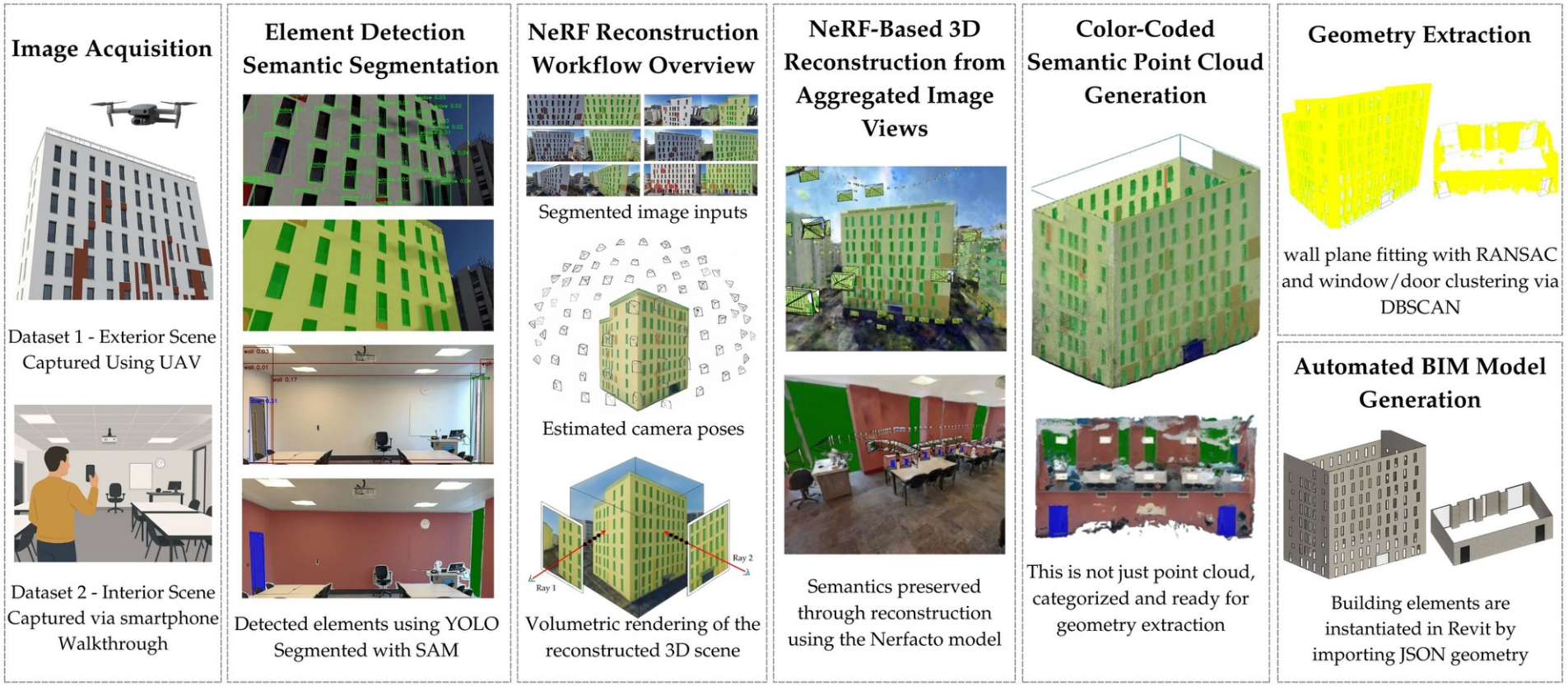
NeRF (Neural Radiance Fields) - Creates 3D scenes from a set of 2D photos.



Examples - 3D Modeling and Animation

NeRF (Neural Radiance Fields) - Creates 3D scenes from a set of 2D photos.

Refs.: Mehraban M.H. et al., Automated Image-to-BIM Using Neural Radiance Fields and Vision-Language Semantic Modeling, Buildings 2025, 15(24), 4549



Rendering with AI

- AI denoising reduces grain in rendered images
- Super-resolution increases image quality without extra computation
- Real-time rendering engines integrate AI to boost frame rates and detail

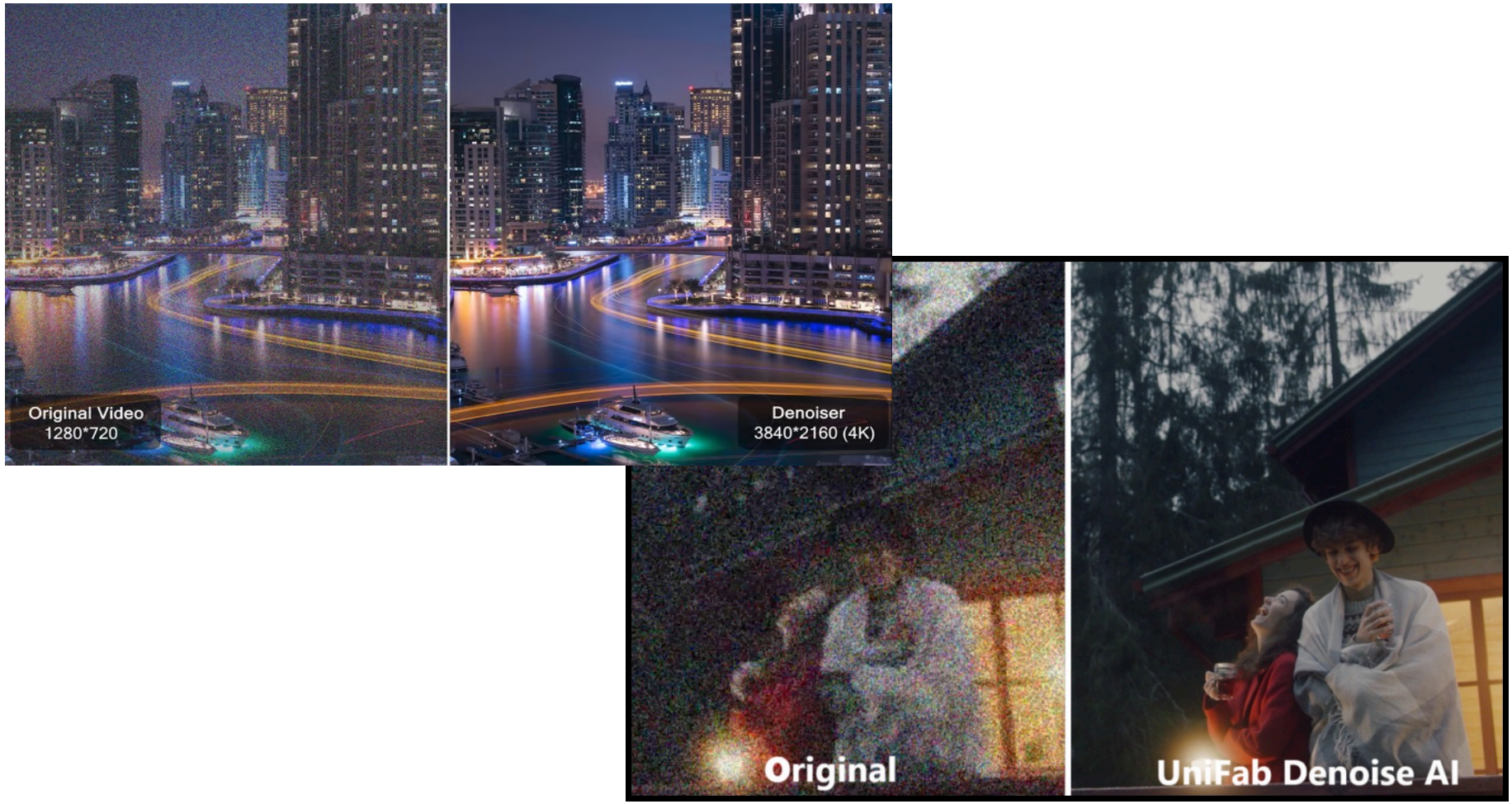
Examples – Rendering with AI

NVIDIA DLSS (Deep Learning Super Sampling) is an AI-powered rendering technology that uses dedicated Tensor Cores on GeForce RTX GPUs to boost frame rates while producing sharp, high-quality images. It works by rendering frames at a lower resolution and using AI to upscale them, reducing the rendering load and delivering up to 4X faster performance.



Examples – Rendering with AI

Across Arnold (Autodesk), Cycles (Blender), and V-Ray (Chaos), UniFab faster production times by cleaning up noisy, low-sample images.

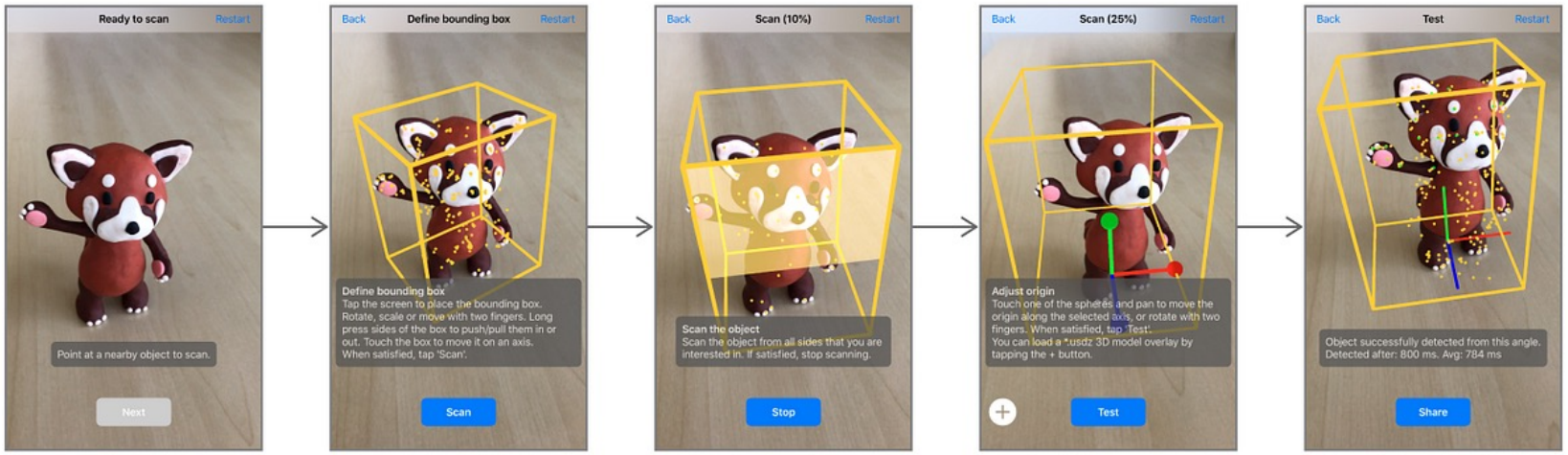


VR and AR Enhancements

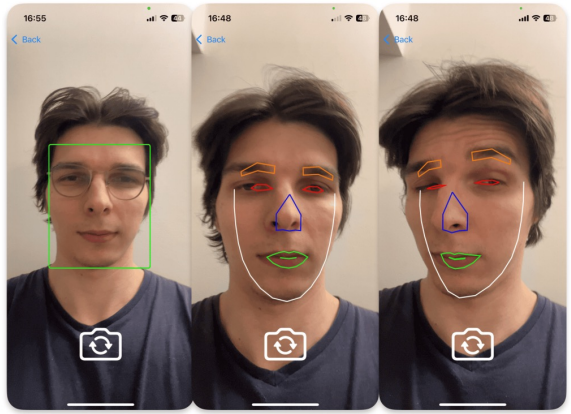
- AI recognizes real-world objects for AR overlays
- Predicts user movement for smoother VR experiences
- Optimizes rendering to maintain high performance on limited hardware

Examples - VR and AR Enhancements

Apple ARKit is one of the dominant mobile AR SDKs enabling 3D object detection, face tracking



Prepare to scan | Define bounding box | Scan | Adjust origin | Test and Export





Examples - VR and AR Enhancements

Google ARCore is one of the dominant mobile AR SDKs enabling 3D object detection, plane detection (horizontal/vertical), and light estimation

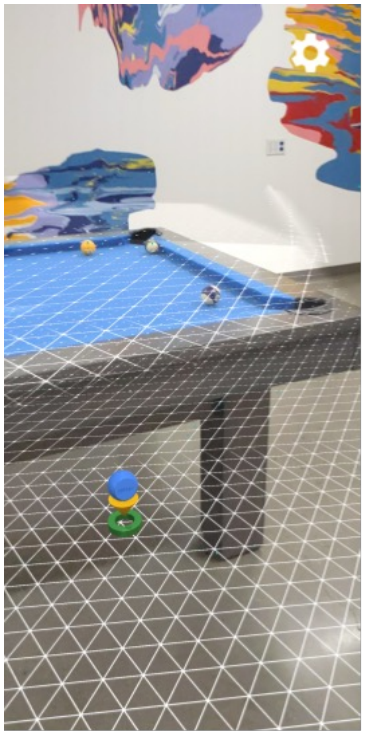


Fig. 1: Place objects without Depth enabled.

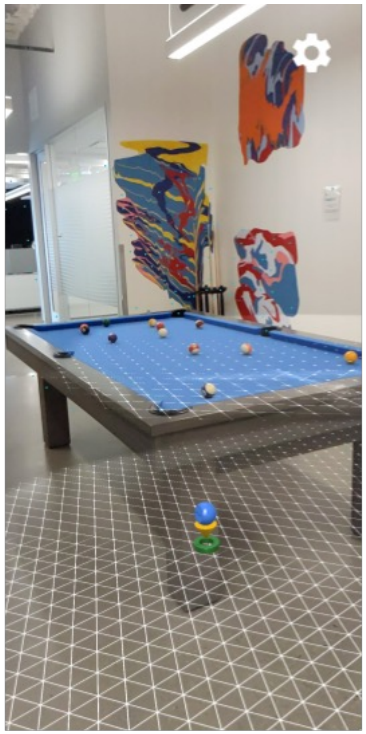


Fig. 2: Object appears irrespective of objects in the scene.

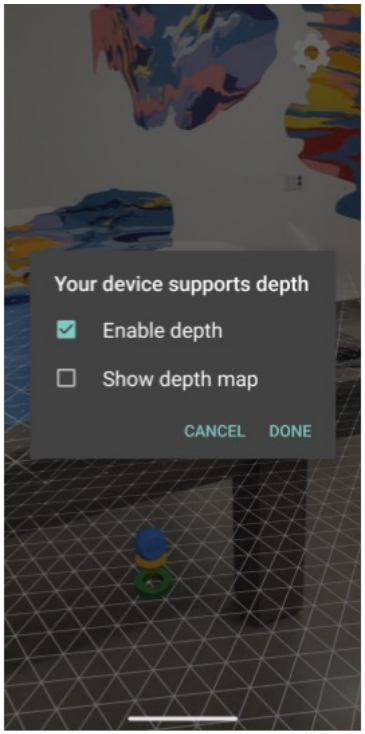


Fig. 3: Enable Depth.

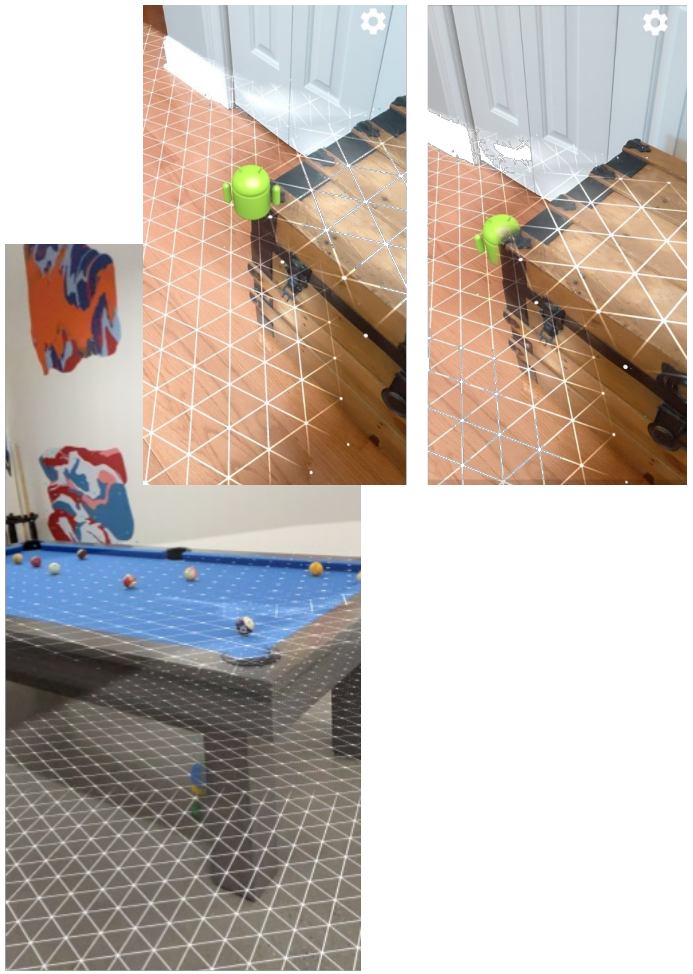
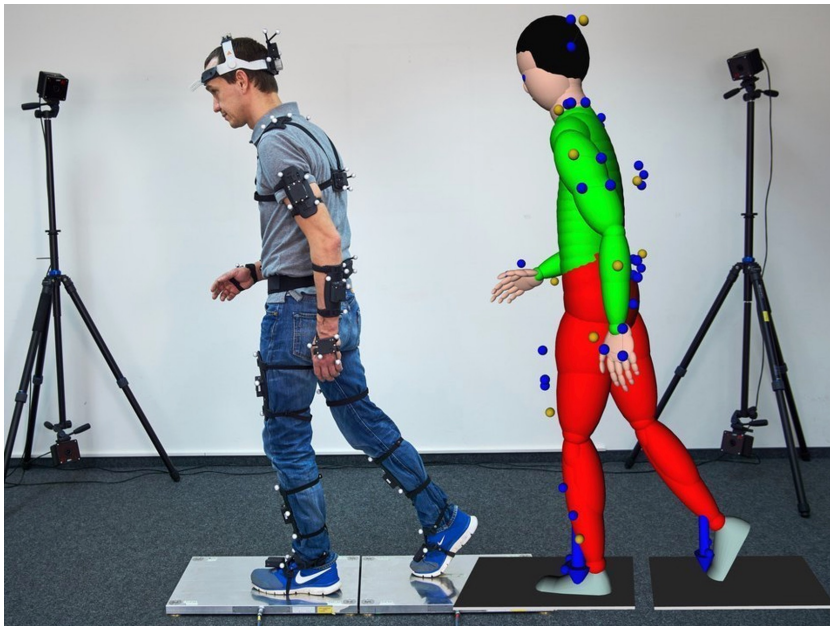


Fig. 4: Placed object is occluded by objects in the scene.

Real-World Examples

- Professional tools: Adobe Photoshop, Blender, Unreal Engine
- AI-generated characters and environments in video games
- Architectural visualization with AI-assisted lighting and materials
- Film production using AI for visual effects and scene generation

Examples – Real World

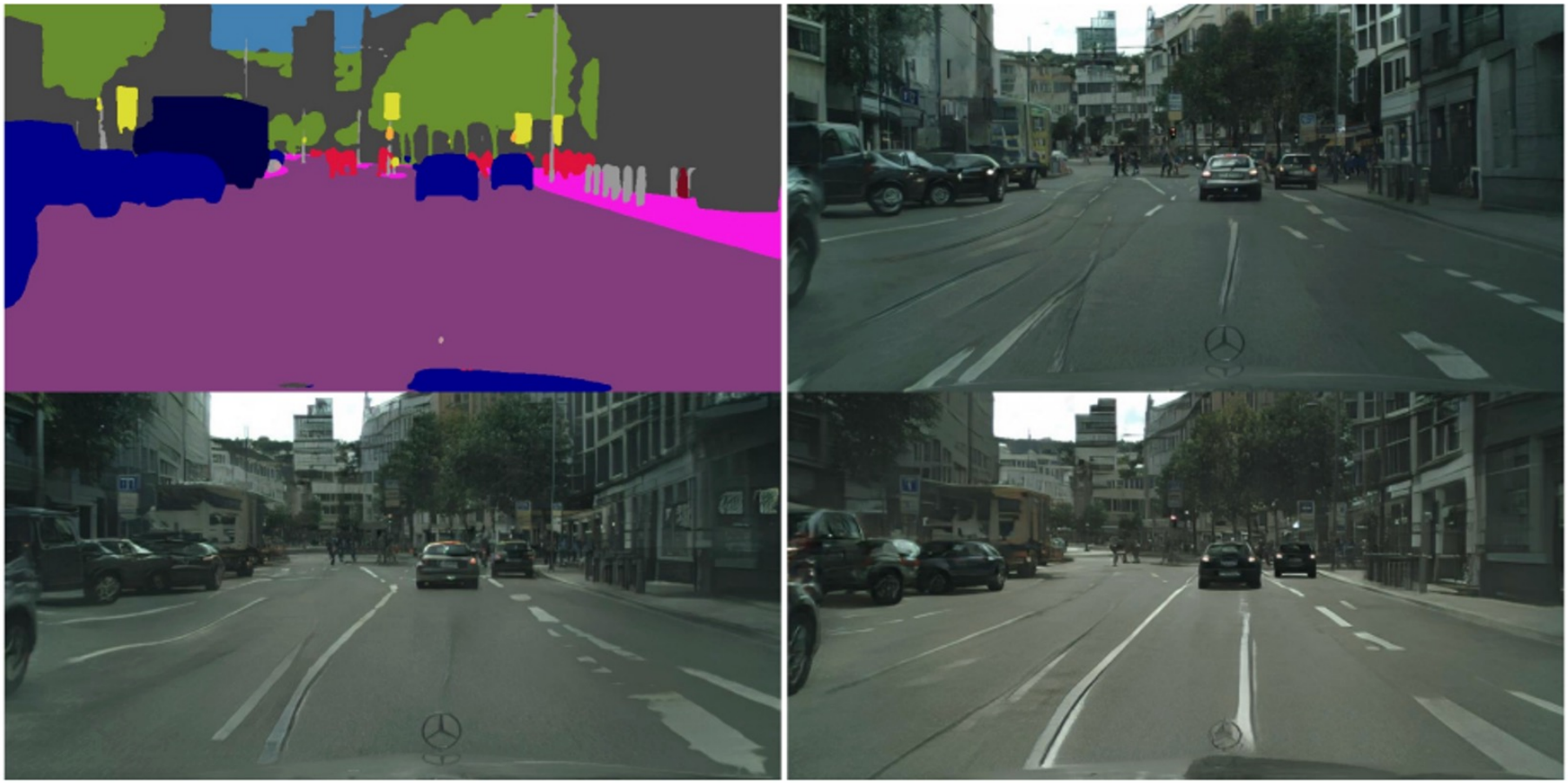






Examples – Real World

Generating a photorealistic video from an input segmentation map vide on Cityscapes. Top left is the input.



Examples – Real World





Examples – Face Replacement

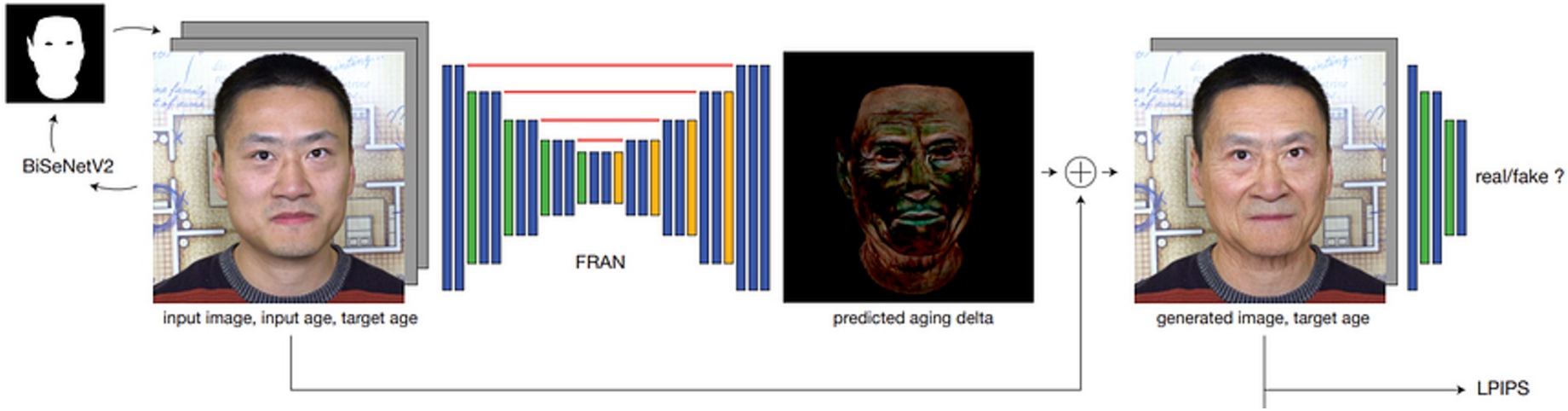
Deepfake-style face replacement, often referred to in professional VFX as AI-powered face swapping or synthetic performance, has matured into a mainstream tool for de-aging actors and replacing stunt doubles.





Examples – FRAN Algorithm

Disney — FRAN (Face Re-Aging Network), AI system that automatically ages or de-ages actors in video.





Examples – Hair Simulation

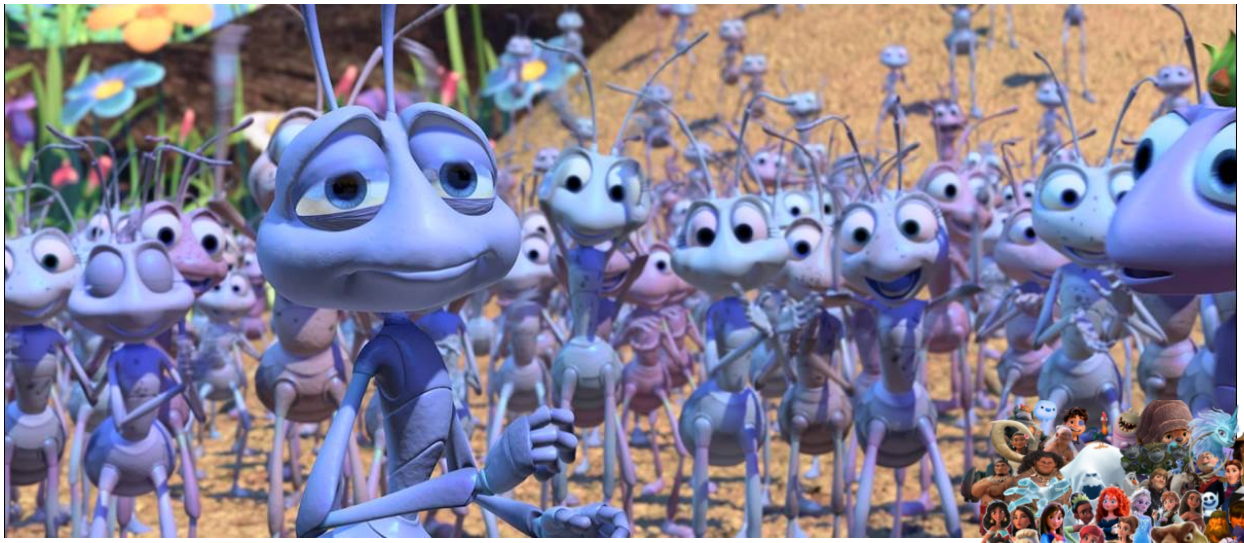
Pixar animate Merida's hair in 'Brave'.





Examples – Crowd Animation

Disney uses machine learning to animate large crowds automatically.







Examples – Cloth Simulation

Pixar uses neural networks to predict how clothing folds and moves. Cloth rigging, autoscaling, and kinematic springs result.





Advantages of AI in Graphics

- Faster workflows
- Higher-quality visuals
- Reduced manual labor
- More creative freedom
- Accessible tools for beginners and professionals alike

Challenges and Limitations

- High hardware requirements
- Need for large datasets
- Ethical concerns: originality, copyright, authenticity
- Limited control over AI-generated results
- Potential for misuse or overreliance

Future Trends

- Full scene generation from text descriptions
- AI-driven real-time rendering in games and simulations
- Personalized visual content
- Fully automated production pipelines
- Integration of AI into every major graphics tool

Conclusion

- AI is reshaping computer graphics by making it faster, smarter, and more creative.
- It enhances both the artistic and technical sides of visual production.
- As AI continues to evolve, it will become an essential part of how we design, animate, and visualize the digital world.